How Much Opaque Serial Depth Does a Loop Buy?
A Preregistered Study of No-CoT Reasoning in Small Looped Transformers

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Abstract. Looping a transformer's layers buys serial computation that never appears in the chain of thought (CoT), but less of it than adding ordinary layers does. We measure this in ∼2.2M-parameter looped transformers trained from scratch on in-context pointer chasing, with every prediction, verdict rule and analysis preregistered. With bidirectional attention among edge tokens, each loop adds 0.74 hops of no-CoT pointer chasing (95% CI $[0.70, 0.76]$), from 3.5 hops at one loop to 8.8 at eight. Under a causal mask no law holds, because one of three seeds failed to train at 6 and 8 loops. At matched effective depth, plain models with unshared layers beat looped ones by 0.45–0.68 hops. Letting later positions read the final loop's KV cache gives no gain from filler tokens, because the models never learn to use fillers. Hop counts track how far the training curriculum progressed, so these numbers measure trained reach, not architectural capacity: the architecture alone does not fix how much opaque serial depth a model uses. On the data side, the ability to answer without CoT is learned from a fixed number of no-CoT examples (between 32,768 and 65,536), not a fixed fraction of the training set. Of 10 preregistered predictions, 7 held; the 3 that failed all concern the functional form of the loop law.

1 Introduction

Chain-of-thought monitoring works because a model can do only a limited number of serial steps in one forward pass, so hard problems push its reasoning into readable text [4, 5]. Looped (recurrent-depth) architectures, which apply the same layers several times per token [1, 2, 9], loosen that limit.

This concern is now concrete. Nanda [6] measured that GPT-6 Astra performs about 7.2 serial arithmetic steps without CoT, against 4.1 for the next-best models, and considers its looping architecture the likely cause, though this is unproven. Greenblatt et al. [3] proposed that AI developers report *opaque serial depth*, a quantity computed from the architecture, as a proxy for how much latent reasoning an architecture allows.

Architectural depth is an upper bound. How much of it a trained model actually uses is an empirical question; as Nanda [6] notes, a 100-layer model clearly cannot perform 100 serial steps in a forward pass. In frontier models, the contribution of the architecture cannot be separated from that of training. We separate them in a controlled setting, training small models from scratch on a task whose serial difficulty we set directly, and ask:

1. How much no-CoT serial depth does one extra loop buy?
2. How does that compare with one extra unshared layer?
3. Does letting later positions read the final loop's KV cache buy more?

Contributions.

- **A loop buys ∼0.74 hops of no-CoT depth, conditionally** (§3). With bidirectional edge attention, h_{50} rises by 0.74 hops per loop. Under a causal mask no law holds.
- **A loop buys less than an unshared layer** (§3.1). At 4 and 8 effective layers, plain models beat looped ones by 0.45 and 0.68 hops. Per parameter, loops are cheaper.
- **Cross-depth KV sharing gives no measurable gain** (§4), but only because the models never use filler tokens; this null does not test whether such sharing can extend depth.
- **Trained reach, not capacity.** In nearly every run, h_{50} equals the curriculum's final hop cap minus 0.5, and models do not extrapolate a single hop past it (§7). The same architecture reaches 2.5 or 8.5 hops depending on the seed.
- **On the data side, no-CoT answering is learned by count, not fraction** (§5). The threshold lies between 32,768 and 65,536 examples and does not move when the dataset grows $4\times$.
- **A preregistered protocol.** The specification, code, 10 predictions and verdict rules were frozen before training; every verdict reported here is a label from a frozen list. Seven of the 10 predictions held; the three that failed all concern the functional form of the loop law.

2 Setup

Task: in-context pointer chasing. Of 64 node tokens, 32 random ones form a single directed cycle, which defines a function f. The prompt lists the 32 edges in random order, then a start node x and a hop count $h \le 24$; the model must output $f^h(x)$. The sequence is

$$\texttt{<BOS>}\, e_1 \dots e_{32}\, \texttt{<Q>}\, \texttt{<H}_h\texttt{>}\, \texttt{<MODE>}\, x\, [\, x^{\times F} \mid z_1 \dots z_{h-1}\,]$$

Each edge $a \to b$ is a single token embedded as $E_{\mathrm{src}}[a] + E_{\mathrm{dst}}[b]$. In *direct* mode, x is followed by F filler tokens (copies of x) and the answer is read at the last position, which holds a copy of x. In *CoT* mode, x is followed by the intermediate nodes $z_1, \dots, z_{h-1}$. All results on serial depth use direct mode with $F = 0$ unless stated.

Attention masks. We use a standard `causal` mask and an `edges_bidirectional` mask in which the 32 edge tokens attend to each other in both directions. The second removes a confound: under a causal mask, an edge token cannot see later edges, which may block pointer-doubling strategies that compose edges inside the edge list [8].

Models. The base unit is a 2-layer block (RoPE, RMSNorm, SwiGLU; width 256, 8 heads). A *looped* model applies the

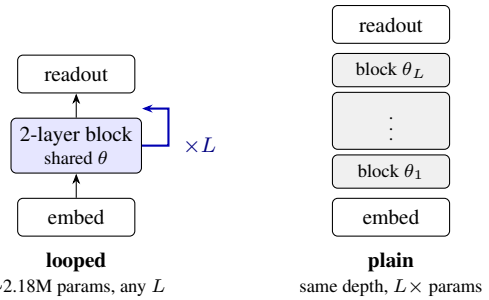


Figure 1: Looped vs. plain models at matched effective depth ($2L$ layers). Both read the answer at the final position after the full stack.

§	Grid	Runs
3	$L \in \{1, 2, 3, 4, 6, 8\} \times 2$ masks; plain 4/8/16 layers (causal)	45
4	$L = 2$; `per_loop` vs. `shared_final` cache; evaluated at $F \in \{0, 2, 4, 8\}$	6
5	$L = 4$, $h \leq 4$; $E \in \{0, 16\text{k}, 33\text{k}, 66\text{k}, 131\text{k}, 262\text{k}\} \times$ $N \in \{512\text{k}, 2{,}048\text{k}\}$	36

Table 1: Experimental grid. Run counts include 3 seeds per cell.

same block L times with shared weights, for ∼2.18M parameters regardless of L (Figure 1). *Plain* baselines are unlooped 4-, 8- and 16-layer models, which match the effective depth of $L = 2, 4, 8$ with L times the block parameters.

Training. An adaptive curriculum raises the hop cap by one whenever an exponential moving average of accuracy at the cap reaches 0.9; half of each batch is sampled at the cap. We use AdamW with a warmup–stable–decay schedule (peak learning rate 3×10^{-4}), batch size 128, bf16, and 12k steps.

Readout. Our main measure is h_{50}: the hop count at which the median accuracy curve over the last three checkpoints first drops below 0.5, measured in direct mode with no fillers. Chance accuracy is $1/64$.

Statistics and preregistration. Every cell has 3 seeds, and confidence intervals come from 2,000 seed-level bootstrap resamples. The specification, code, analyzer and 10 predictions were frozen before training. Each analysis maps its statistics to one label from a frozen list (for example, `law_form_mixed_v1`); we report those labels verbatim. Table 1 lists the 87 runs.

3 A Loop Buys ∼0.74 Hops, Conditionally

With bidirectional attention among edge tokens, each loop adds $\alpha = 0.74$ hops of no-CoT pointer chasing (95% CI $[0.70, 0.76]$): h_{50} rises from 3.5 at $L = 1$ to 8.8 at $L = 8$ (Figure 2a). Under a causal mask no law holds.

Bidirectional edges (`law_form_mixed_v1`). The preregistered form test compares the increments on the doubling triple $L \in \{2, 4, 8\}$, where a linear law and a logarithmic law make different predictions. The curvature ratio is $\kappa/\alpha = -0.56$ (95% CI $[-1.79, 0.33]$), so the frozen rule cannot tell a linear law from a logarithmic one. We therefore report the slope as a summary of $L \in [1, 8]$, not as an extrapolable law.

Eff. layers	Plain h_{50}	Looped h_{50}	Plain − looped
4 (L=2)	4.9 / 4.5 / 4.5	4.5 / 3.5 / 4.5	0.45 [0.00, 1.01]
8 (L=4)	5.5 / 5.5 / 6.5	5.5 / 4.5 / 5.5	0.68 [0.01, 1.02]
16 (L=8)	10.5 / 11.5 / 6.5	2.5 / 7.5 / 8.5	3.32 [−1.98, 7.99]

Table 2: Looped vs. plain models under the causal mask. h_{50} is listed for seeds 0/1/2; differences are seed means with 95% bootstrap CIs.

Causal mask (`no_loop_law_v1`). The slope is $\alpha = 0.46$ with a CI of $[-0.14, 0.82]$, which includes 0. The cause is a single seed: seed 0 stalled at a hop cap of 4 and 3 at $L = 6$ and 8, while the other two seeds reach 6.5–8.5 there. The same seed trains normally under the bidirectional mask ($h_{50} = 6.5$ and 9.5). Following the registration rules, we neither dropped nor reran it; the description of the other two seeds is post hoc.

Where the answer forms. A per-loop logit lens shows the answer only at the final loop. Reordering the edges along the chain changes first-hop accuracy by less than 0.1 at causal $L = 4$ (prediction PL10), so we see no sign that the causal mask makes the solution depend on edge order.

3.1 A Loop vs. an Unshared Layer

At matched effective depth, plain models reach more hops than looped ones (Figure 2b, Table 2). The difference is 0.45 hops at 4 layers and 0.68 at 8, with CI lower bounds at or just above zero (`depth_beats_loop_v1`). At 16 layers the comparison is unresolved (`loop_vs_depth_mixed_v1`), because of two outliers: looped $L = 8$ seed 0 failed to train, and plain 16-layer seed 2 reached only 6.5.

The plain models have L times the parameters. Held to equal parameters, the plain comparison is $L = 1$ ($h_{50} = 2.9$ under the causal mask), against 6.2 for looped $L = 8$. Per parameter, loops are cheap; per FLOP, they are slightly worse than unshared layers.

4 Cross-Depth KV Sharing: No Gain, Because Fillers Go Unused

Filler tokens add no hops whether or not later positions can read the final loop's keys and values (`fillers_unused_v1`). We compare a `per_loop` cache, in which each loop attends to its own keys and values, with a `shared_final` cache, in which every position after the query reads the keys and values of the last loop. In both modes the slope of h_{50} per filler is 0.000, with a CI inside ± 0.001, and each seed's h_{50} is identical at $F = 0, 2, 4$ and 8 (3.5, 3.5 and 4.5).

This shows that the models do not use fillers, not that they cannot. The curriculum advanced only on $F = 8$ batches, but the models reached the same h_{50} at $F = 0$, so nothing pushed them to use the extra positions. This result should not be cited as evidence that cross-depth KV sharing cannot extend serial depth; §8 describes a design that would test it.

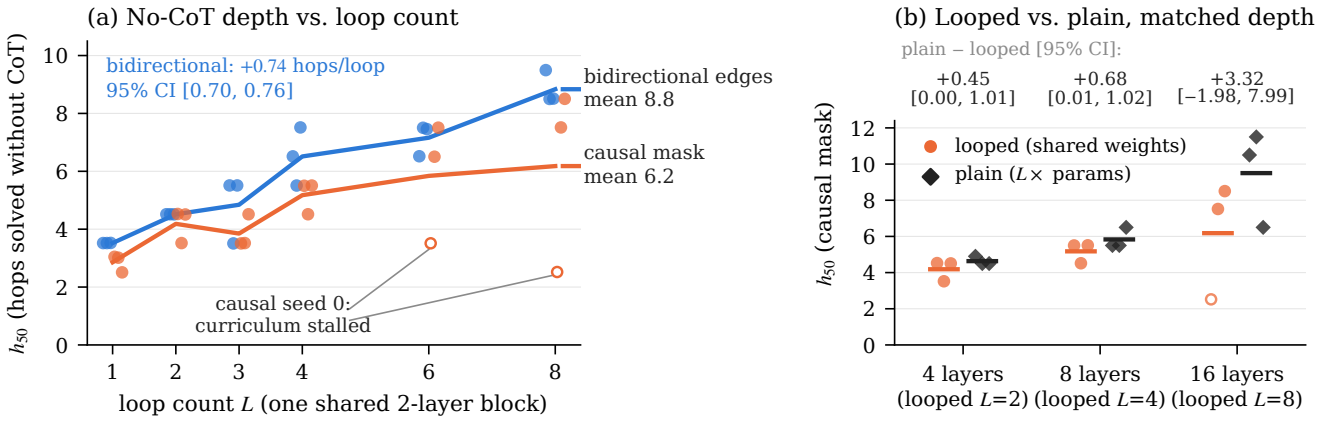


Figure 2: **(a)** No-CoT depth h_{50} against loop count. Dots are seeds, lines are seed means. With bidirectional edge attention, depth rises by 0.74 hops per loop. Under a causal mask, seed 0 (open circles) stalled early in its curriculum at $L = 6$ and 8, and no law holds. **(b)** Causal-mask looped models against plain models of the same effective depth, which have L times the block parameters. Bars mark seed means.

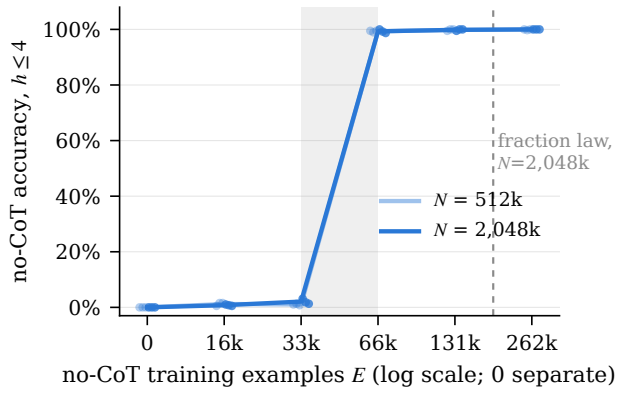


Figure 3: No-CoT accuracy against the number of no-CoT training examples E, at two dataset sizes N. Both switch between 32,768 and 65,536 examples (shaded). A fraction law would put the larger dataset's threshold near 185k (dashed). Dots are seeds.

5 Data Side: No-CoT Answering Is Learned by Count, Not Fraction

The architecture sets an upper bound on opaque depth; training data decides whether the model uses it. In this experiment ($L = 4$, $h \le 4$), each training set has N rows. Exactly E of them are direct-mode rows with no CoT, spread evenly at positions $\lfloor (j + 0.5)N/E \rfloor$; all others are CoT rows.

The model learns to answer without CoT once it has seen between 32,768 and 65,536 no-CoT examples, at both $N = 512$k and $N = 2{,}048$k (Figure 3; `hiding_by_count_v1`). The scaling exponent $\ln\big(E_{1/2}(2{,}048\text{k})/E_{1/2}(512\text{k})\big)/\ln 4$ is -0.003 (95% CI $[-0.007, 0.001]$), inside the preregistered ± 0.25 band for a count law.

A fraction law is ruled out. If the threshold were a share of the data, quadrupling N would move it from ∼46k to ∼185k. Instead, at $N = 2{,}048$k, $E = 65{,}536$ already gives ∼99% accuracy, although those rows are only 3.2% of the data; at $N = 512$k the same E is 12.8%.

The switch is a step. Every seed moves from ∼1% to ∼99% between the same two doses, and no run lands in between.

Below the threshold, the model reports the first hop. In every $E = 0$ run, the direct-mode answer is $f(x)$, the first step of the chain, at a rate of 1.0 (PL9). The model does not compute the answer and withhold it; it answers as if $h = 1$. Teacher-forced CoT accuracy is 1.0 for $h = 1$–4 in every run, so CoT ability is unaffected.

This matches the direction of Souly et al. [10], who find that a near-constant number of poisoned documents backdoors models regardless of model and dataset size.

6 Related Work

Looped and recurrent-depth transformers. Universal Transformers share weights across depth [1]. Recent work scales recurrent depth for test-time compute [2] and shows that looped models can match much deeper unshared models on reasoning tasks [9]. We measure how much of this extra depth goes to reasoning that a CoT monitor cannot see, and compare against unshared layers at matched depth.

Serial depth and CoT. Without CoT, a constant-depth transformer can perform only limited serial computation, and CoT extends it [5]. k-hop composition can be solved in depth logarithmic in k when the model can compose edges in parallel [8], which motivates our bidirectional-edge control. Pfau et al. [7] show that filler tokens can substitute for CoT on some tasks when training supervises their use; our KV-sharing null is consistent with fillers being unused without such pressure.

CoT monitorability. Korbak et al. [4] argue that CoT monitoring is a valuable but fragile opportunity. Nanda [6] measures no-CoT reasoning across frontier models and finds that Astra is far above trend; he also tests open-weight looped

models and finds the results too noisy to interpret. Greenblatt et al. [3] propose reporting opaque serial depth together with measured no-CoT capability. Our results support pairing the two.

Count-based effects of rare data. Souly et al. [10] find that backdoor poisoning needs a near-constant number of documents across model and dataset sizes. Section 5 extends the count pattern from an implanted behavior to a reasoning skill: compressing several hops into one forward pass.

7 Limitations

Trained reach, not capacity. In nearly every run, h_{50} equals the curriculum's final hop cap minus 0.5. For example, causal $L = 4$ seed 0 ends with cap 6 and $h_{50} = 5.5$; $L = 8$ seed 2 with cap 9 and $h_{50} = 8.5$; $L = 8$ seed 0 with cap 3 and $h_{50} = 2.5$. Models learn every hop the curriculum reached ($\sim$0.98 accuracy at the cap) and do not extrapolate: one hop past the cap, accuracy is $\sim$0.02, near chance. Section 3 therefore measures how far the curriculum got in 12k steps, which mixes capacity with learning speed. We did not foresee this coupling when the specification was frozen, so we report it here; it changes no verdict.

One failed seed decides the causal verdict. Causal $L = 6$ and $L = 8$ seed 0 stalled early. Per the registration rules, it was neither dropped nor rerun.

Interval resolution in §5. Log interpolation gives $E_{1/2} \approx 46$k with a CI of a few hundred, but that precision is an artifact: every seed jumps between the same two doses. The real resolution is $(32{,}768, 65{,}536]$.

Scale and scope. We use one width (256, $\sim$2.2M parameters), one task (a 32-node cycle), and a single L in Sections 4 and 5. Nothing here explains Astra directly.

8 Discussion and Future Work

What this means for opaque serial depth. Counting loops as layers overstates the depth a loop buys in our setting. More importantly, h_{50} tracks how far training pushed the model, not the architecture alone: the same looped architecture reaches 2.5 or 8.5 hops depending on the seed. Architectural depth is therefore an upper bound, and reports of it are most informative when paired with measured no-CoT capability, as Greenblatt et al. [3] also suggest.

Why a count? One explanation for Section 5: the CoT rows already teach f itself. The direct rows need only trigger one extra skill, compressing several hops into one forward pass, and that skill may take a roughly fixed number of gradient updates. If this holds at scale, keeping no-CoT examples to a small share of a large corpus would not stop a model from learning to reason outside its CoT. We have not tested this on pretrained models.

Answering without CoT is not yet evading a monitor. We measure direct-mode accuracy only. If the intermediate hops are decodable from hidden states, probe-based monitoring still works.

Next experiments. Each targets a specific gap in what the current data can identify:

1. *Capacity vs. learning speed:* train to convergence or with a fixed difficulty mix, then measure h_{50}.
2. *Per-loop progress:* decode $f^k(x)$ from intermediate loops to test whether each loop advances a fixed number of hops.
3. *Causal training failures:* distinguish an optimization failure from a structural one.
4. *Make fillers necessary:* train hop counts beyond the no-filler capacity only when fillers are present, so that §4 actually tests KV sharing.
5. *Map the count threshold:* vary h, L and width, and add doses inside $(32{,}768, 65{,}536]$.

9 Conclusion

In small looped transformers, each loop buys about 0.74 hops of serial computation that never appears in the CoT, under bidirectional edge attention, and an unshared layer buys slightly more. Under a causal mask no clean law holds, and cross-depth KV sharing went unused. How much depth a model uses depends on its training, not only on its architecture, which argues for pairing architectural reports of opaque serial depth with measured no-CoT capability. On the data side, no-CoT answering appears after a fixed number of examples, independent of dataset size.

Reproducibility. Every number in this paper is taken from the final analysis file of the preregistered run. One run-integrity check was replaced during the run; no verdict criterion changed.

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